A Retrospective on Iterated Expectation

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In general, thanks to iterated expectation—also known as Adam's Law—the **unconditional expectation of a conditional expectation is** simply the **unconditional expectation**:

$$\mathbf{E}[\,\mathbf{E}[X \mid U, V, \ldots]\,] = \mathbf{E}[X].$$

The **conditional expectation of a conditional expectation behaves like a filter**: whenever the outer conditioning set is a subset of the inner, the result collapses to the outer conditional expectation. For example,

$$\mathbf{E}\Big[\mathbf{E}[X \mid U, V, Y, Z] \,\Big|\, V, Z\Big] \;=\; \mathbf{E}[X \mid V, Z].$$

If the **conditioning set contains the random variable** itself, the expectation **collapses** to that variable. For example,

$$\mathbf{E}[X \mid W, X, Y] \;=\; X.$$

Trivially, any **expectation**—conditional or unconditional—of an **unconditional expectation is** just **the unconditional expectation**, simply because $\mathbf{E}[X]$ is a constant and therefore independent of any conditioning information:

$$\mathbf{E}\Big[\mathbf{E}[X] \,\Big|\, X, Y, Z\Big] \;=\; \mathbf{E}[X].$$

It is also worth pointing out the immense power of iterated expectation—it provides a shortcut to complicated problems, bypassing integration (or summation) entirely, and is especially valuable when no closed form is available.

Happy iterating expectations!

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